

10(a)	<i>any six from:</i> 1 analysis of biological data using computer software / AW ; 2 databases of, gene / DNA / protein / amino acid, sequences ; 3 ref. to large databases ; 4 fast / accurate / efficient ; 5 ref. to allows data to be, shared / pooled ; 6 can predict, amino acid sequences / protein structure (from DNA sequence data) ; 7 ref. to analytical tool ; e.g. BLAST 8 ref. to comparisons ; 9 used to find methods to control parasites ; 10 named example of control ; e.g. vaccine 11 AVP ; e.g. personalised medicine identify new diseases ref. common ancestor phylogenetic analysis biodiversity / new species	6
-------	--	----------

Question	Answer	Marks
10(b)	any nine from: 1 probes are, single-stranded DNA / ssDNA ; 2 each probe is unique to a particular gene ; 3 probes correspond to thousands of different genes ; 4 extract mRNA ; 5 mRNA used (as a template) to make <u>c</u>DNA ; R mRNA converted to cDNA 6 by reverse transcription / using reverse transcriptase ; 7 (c)DNA linked to fluorescent dye ; 8 (c)DNA added to microarray ; 9 (c)DNA, binds to / hybridises, to probes ; 10 by complementary base pairing ; 11 excess (c)DNA washed off ; 12 exposed to, UV light / laser ; 13 fluorescence shows the expressed genes / AW ; 14 intensity of fluorescence shows level of gene expression ;	9